地政總署
Lands Department

AECOM

Simplified Data Specifications for

BUILDING FSDT



Version 1.1
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Foreword

To promote better sharing and accessibility of spatial data among the community of stakeholders in Hong Kong Special Administrative Region ("HKSAR"), the Government acknowledged its commitment to establishing Common Spatial Data Infrastructure ("CSDI"). The main objective is to provide Bureaux/Departments ("B/Ds") and public and private organisations with an information infrastructure to share spatial data, support various smart city applications, and support Smart City development.

In March 2017, the Development Bureau ("DEVB") commissioned a consultancy study on the overall strategy for implementing the CSDI. With its implementation, the CSDI would benefit the Government and society in the long run, with convenient and easy access to high-quality and up-to-date spatial information. The CSDI will help to:

- Facilitate and support the development of a dynamic and flexible geospatial environment that will provide standardised interfaces for accessing and disseminating common spatial information;
- Facilitate information sharing across different B/Ds and eventually other stakeholders;
- Encourage the adoption of common standards to ensure quality, compatibility and interoperability amongst different systems; and
- Make available a wide range of spatial data and analyses to enable potential applications.

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1. Introduction

1.1. General Overview

The Simplified Data Specifications for Building ("the/this document") provide information about the datasets in the Building Framework Spatial Data Theme ("FSDT").

1.2. Glossary and Definitions

This section provides a glossary of terms and definitions, listed alphabetically, as applicable to this document.

- **Feature:** Abstraction of real-world phenomena. A feature may occur as a type or an instance.
- **Feature Attribute:** Characteristic of a feature. A feature attribute has a name, a data type, and a value domain associated to it.
- **Polygon:** Topological 2-dimensional geometric primitive, representing an areal feature.

1.3. Acronyms and Abbreviations

This section lists the expanded description of all acronyms (words formed from the initial letters of a phrase) and abbreviations (the shortened form of a word or phrase) used in this document in alphabetical order.

Acronym/Abbreviation	Description
API	Application Programming Interface
BD	Buildings Department
B/D	Bureau/Department
BCIS	Building Condition Information System
CSDI	Common Spatial Data Infrastructure
CSU	Common Spatial Unit
CSUID	Common Spatial Unit Identifier
DAM	Data Alignment Measures
DEVB	Development Bureau

FBX	Filmbox
FSDT	Framework Spatial Data Theme
glTF	Graphics Library Transmission Format
HKPD	Hong Kong Principal Datum
HKSAR	Hong Kong Special Administrative Region
KML	Keyhole Markup Language
LandsD	Lands Department
MAX	Autodesk 3ds Max
OmniClass™ / OCCS	OmniClass Construction Classification System
OSGB	OpenSceneGraph Binary
SI	International System of Units
the/this document	Simplified Data Specifications for Building
URL	Uniform Resource Locator
UUID	Universally Unique Identifier

2. Framework Spatial Data Definitions

2.1. Overview

The Building FSDT covers building structures that enclose spaces including residential, commercial, industrial, infrastructure, government, institution or community purposes.

2.2. Scope & Definitions

This document for Building covers the following sub-themes:

- **Building Outline:** Building outlines are features delineating the outline of structures that are visible above-ground. Based on existing definitions from DAM Building Common Spatial Unit ("CSU"), building outlines are classified as either Tower or Podium. Towers are defined as all permanent buildings, open side and temporary structures of a reasonable size as stipulated in the latest 1:1000 Basic Mapping Specifications prepared by Lands Department ("LandsD"), while podium is defined as containment with facilities for a designated purpose (e.g. shops, car park etc.) above ground. Usually, podium together with the tower would form the integral building structure.

  To align with Smart City and other strategic innovation initiatives, underground building structure[1] which include any underground storey of a tower/podium below the ground storey with an exit route is also scoped to be covered. The underground extent shall be delineated by the maximum extent of all floors, modelled as one or more level, that exist below ground, as demarcated in building plans.

- **Building Shell:** The building shell is based on all the enclosures that represent the exterior of the entire building structure, both above the ground and below. It includes all sub-parts of a building, such as parts of a building that have different roof heights or multiple towers on a single podium.

- **Floor:** Floors represent the different levels, or stories, within a building. Floors are inclusive of units, common areas, stairwells, and other spaces comprising the entire area of a single level, or storey.

[1] The definition, spatial delineation and examples of underground structure in the Building Outline sub-theme will be further reviewed upon the readiness of data from B/Ds. The limitations and improvements regarding the inclusion of underground structures would be documented in Final Report of the Feasibility and Technical Study on Framework Spatial Data Standards for the Development of Common Spatial Data Infrastructure (CSDI) for Lands Department.

- **Unit[2]:** Unit, in relation to a building, means premises of the building falling within either or both of the following descriptions-

    a) premises that are demarcated or shown as a separate unit (however described) in the approved building plan of the building (whether domestic or non-domestic);

    b) premises that the owner of which is entitled to its exclusive possession, as opposed to the owners or occupiers of other parts of the building.

- **Room:** Room means any portion of a building which has been physically bounded and separated from other spaces, usually by walls or doors.

[2] The definition of Unit may be subject to further review upon the availability of data from B/Ds.

3. Framework Spatial Data Content & Structure

3.1. Datasets under Sub-themes

There are five sub-themes within the Building FSDT, namely Building Outline, Building Shell, Floor, Unit, and Room.

Building Outline sub-theme covers the following dataset(s):

- Building: A polygon showing the building outline of permanent buildings, including tower and podium, temporary structures or open-sided structures. Attributes include the building types, building names, building height etc.

Building Shell sub-theme covers the following dataset(s):

- 3D Visualisation Map (Individualised models): The 3D Visualisation Map (Individualised models) are a set of digital data of 3D models featuring geometry models and texture maps to represent the geometrical shape, appearance and position of different types of ground objects, including building, infrastructure, vegetation, site, waterbody, terrain and generic (others).
- 3D Visualisation Map (Tile-based models): The 3D Visualisation Map (Tile-based models) are based on the mesh model made from the oblique aerial images.
- 3D Visualisation Map (Non-textured models): The 3D Visualisation Map (Non-textured Models) is a set of digital data of 3D models featuring geometric models to represent the geometrical shape and position of building, infrastructure and terrain.

Floor sub-theme would cover the following dataset(s):

- Floor[3]: A polygon showing the floor inside the building or structure. Floors represent the different levels, or stories, within a building. Floors are inclusive of units, common areas, stairwells, and other spaces comprising the entire area of a single level, or storey.

Unit sub-theme does not contain any dataset[4].

Room sub-theme would cover the following dataset(s):

[3] When dataset(s) become available under this sub-theme, the specification would be further reviewed to incorporate the data specifications of the available dataset(s).

[4] Although Unit sub-theme does not contain any dataset so far, such "Unit" sub-theme must be kept to fit for the operational needs of different B/Ds and to avoid unsurmountable effort of data conversion or data matching. Should Unit data be available in future (if any), the specification would be further reviewed to incorporate the Unit specifications.

- Room: A polygon showing the room inside the building or structure. Room means any portion of a building which has been physically bounded and separated from other spaces, usually by walls or doors.

3.2. Physical Data Model

This section describes the physical data model for each sub-theme within the Building FSDT.

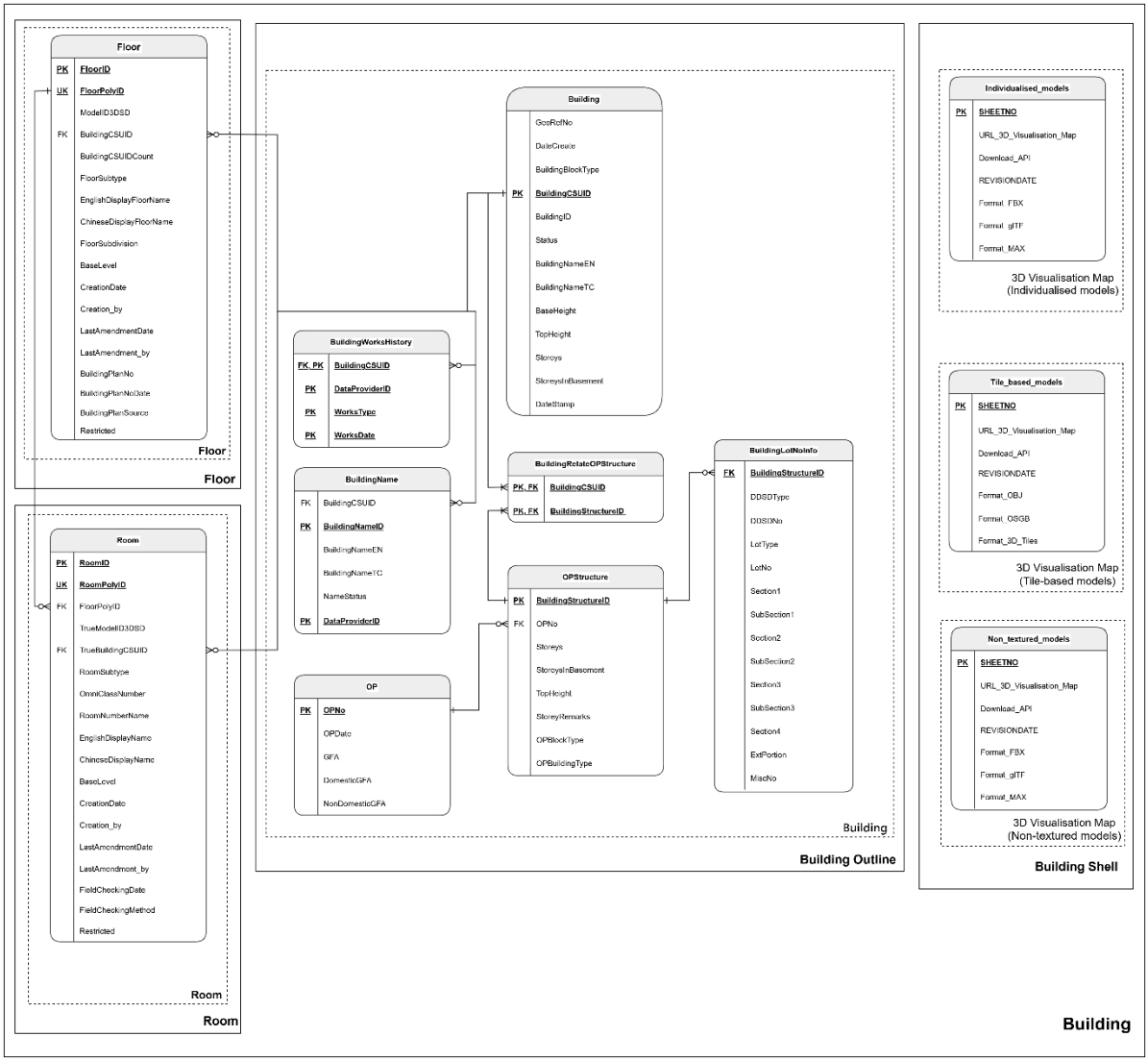


Figure 3-1 Physical Data Model of Building FSDT

3.3. Entity Description

This section provides a detailed description of each physical entity illustrated in the physical data model in Section 3.2.

3.3.1. Building Outline - Building

3.3.1.1. Building

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
GeoRefNo	String, 10	Geo-reference No. of the label point of the building polygon	N
DateCreate	Date	Date on which the building polygon is first created	N
BuildingBlockType	String, 20	Type of the building polygon (e.g. “Tower”, “Podium”, “Open-sided Structure”, “Temporary Structure”)	N
BuildingCSUID	String, 19	Building Common Spatial Unit (CSU) Identifier (e.g. “3012425857T20050430”)	N
BuildingID	Long, 10	Building ID from LandsD’s iB1000 Topographic Map (e.g. “1108234442”)	N
Status	String, 10	Status of Building Structure (e.g. “Proposed”, “Active” or “Demolished”) - Proposed – Status that describes the buildings which are under design/construction; or the buildings of which the construction is completed but the geometry of respective building polygons are yet updated. - Active - Status that describes the buildings which are substantially completed and the geometry of the respective building polygons is available either from land survey or from other verified source. - Demolished – Status that describes the buildings already demolished from Ground.	N
BuildingNameEN	String, 195	Building Name (English) (e.g. “CHAU WAN BUILDING”)	Y
BuildingNameTC	String, 100	Building Name (Traditional Chinese) (e.g. “秋雲大廈”)	Y

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BaseHeight	Double (8,2)	Approximate Level of the base of Building Structure above Hong Kong Principal Datum (in metres) (e.g. “4.3”)	Y
TopHeight	Double (8,2)	Approximate Elevation of the top of Building Structure above Hong Kong Principal Datum (in metres) (e.g. “42.2”)	Y
Storeys	Short, 3	Number of storeys of the building (e.g. “8”)	Y
StoreysInBasement	Short, 3	Number of storeys in the basement (e.g. “8”)	Y
DateStamp	Date	Date when the record is last created/modified (i.e. when new data value(s) from Data Owner(s) is imported into the Building dataset)	N

3.3.1.2. BuildingWorksHistory

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingCSUID	String, 19	Building Common Spatial Unit (CSU) Identifier (e.g. “3012425857T20050430”)	N
DataProviderID	String, 6	Data Provider	N
WorksType	String, 80	Works Type	N
WorksDate	String, 10	Date on which the Works was Carried out	N

3.3.1.3. BuildingName

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingCSUID	String, 19	Building Common Spatial Unit (CSU) Identifier (e.g. “3012425857T20050430”)	N
BuildingNameID	Long, 10	Identifier of the Building Name from Data Provider (e.g. “1109618882”)	N
BuildingNameEN[5]	String, 195	Building Name (English) (e.g. “CHAU WAN BUILDING”)	Y

[5] In Hong Kong, RVD is responsible for the allocation of building numbers throughout Hong Kong and for ensuring the display of correct building numbers, under the Buildings Ordinance (Cap. 123). To avoid confusion to the B/Ds and/or public, only one 2D address record would be provided for each spatial object, and RVD addresses with official building numbers assigned under Cap. 123, if available, must prevail over LandsD surveyed addresses and/or other addresses.

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingNameTC[5]	String, 100	Building Name (Traditional Chinese) (e.g. “秋雲大廈”)	Y
NameStatus	String, 20	Building Name Status (e.g. “Default”, “Other name (alias)”, “Proposed” or “Renamed”)	N
DataProviderID	String, 6	Data Provider	N

3.3.1.4. BuildingRelateOPStructure

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingCSUID	String, 19	Building Common Spatial Unit (CSU) Identifier (e.g. “3012425857T20050430”)	N
BuildingStructureID	Long, 9	Building Structure Identifier (e.g. “15281298”)	N

3.3.1.5. OPStructure

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingStructureID	Long, 9	Building Structure Identifier (e.g. “15281298”)	N
OPNo	String, 15	Occupation Permit Number (e.g. “NT10/2009(OP)”)	Y
Storeys	Short, 3	Number of storeys in podium and above basement. (e.g. “8”) This means the number of above-basement storeys in the podium itself for podium blocks, and means number of above-basement storeys in the podium below for tower blocks.	Y
StoreysInBasement	Short, 3	Number of storeys in basement/lower ground. (e.g. “4”) This counts only the storeys covered by the concerned Building CSU. For example, the basement storeys are counted into the podium block only but not to the towers sitting above.	Y

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
TopHeight	Double (8,2)	Elevation of building above principal datum in metres. Top level of a building is measured to the mean height of the roof over the highest usable floor space in the building. (e.g. “52.3”)	Y
StoreyRemarks	String, 200	Textual description to clarify number of storeys of the building	Y
OPBlockType	String, 6	Building block type as recorded by Building Department (BD) (e.g. “Tower” and “Podium”)	N
OPBuildingType	String, 500	Types of building as of OP (e.g. “Industrial”, “Commercial”)	Y

3.3.1.6. OP

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
OPNo	String, 15	Occupation Permit Number (e.g. “NT10/2009(OP)”)	N
OPDate	Date	Issue Date of Occupation Permit	Y
GFA	Double (10,2)	Gross Floor Area (in sq. metres) The sum of Domestic GFA and Non-domestic GFA (e.g. “990.39”)	Y
DomesticGFA	Double (10,2)	Domestic Gross Floor Area as stated on Building Plan covered by Occupation Permit (e.g. “178600”)	Y
NonDomesticGFA	Double (10,2)	Non-domestic Gross Floor Area as stated on Building Plan covered by Occupation Permit (e.g. “2482.3”)	Y

3.3.1.7. BuildingLotNoInfo

Entity type: Data Table

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BuildingStructureID	Long, 9	Building Structure Identifier (e.g. “15281298”)	N
DDSDType	String, 5	DD/SD Type (English) (e.g. “DD”, “SD”)	Y
DDSDNo	String, 4	DD/SD Number (e.g. “1”, “2”)	Y
LotType	String, 15	Lot Type Abbreviation (English) (e.g. “LOT”, “CLL”, “IL”, “KIL”)	N
LotNo	String, 13	Lot Number (including both numeric and alphabetical part, e.g. “123”, “328”, “20A”)	N
Section1	String, 4	The first section letter (e.g. “A”, “B”, “AA”, “C-F”) appeared in a lot designation. Special values for non-section letters are enclosed in brackets, where: (PE) = PRAYA EXTENSION TO (PR) = PRAYA EXCLAMATION TO (RP) = THE REMAINING PORTION OF (RC) = THE RECLAMATION TO	Y
SubSection1	String, 4	The first subsection number (e.g. “1”, “2”) appeared in a lot designation	Y
Section2	String, 4	The second section letter (e.g. “A”, “B”) appeared in a lot designation.	Y
SubSection2	String, 3	The second subsection number (e.g. “1”, “2”) appeared in a lot designation.	Y
Section3	String, 4	The third section letter (e.g. “A”, “B”) appeared in a lot designation.	Y
SubSection3	String, 3	The third subsection number (e.g. “1”, “2”) appeared in a lot designation.	Y
Section4	String, 4	The fourth section letter (e.g. “A”, “B”) appeared in a lot designation.	Y
ExtPortion	Short, 3	Extended portion, where 1 = and the first extension thereto 2 = and the second extension thereto 3 = and the third extension thereto 4 = and the fourth extension thereto 5 = and the fifth extension thereto 6 = and the sixth extension thereto 7 = and the seventh extension thereto 8 = and the eighth extension thereto 9 = and the extension thereto	Y
MiscNo	Short, 4	Miscellaneous Number of Lot Designation (as in LR’s LRS/IRIS)	Y

3.3.2. Building Shell – 3D Visualisation Map (Individualised Models) (Grid Index)

3.3.2.1. Individualised_models

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
SHEETNO	String,9	iB1000 Sheet No	N
URL_3D_Visualisation_Map	String, 25	URL of 3D Visualisation Map	N
Download_API	String, 35	URL of Download Application Programming Interface (API)	N
REVISIONDATE	Date	Revision Date	N
Format_FBX	String, 120	Path of FBX File	N
Format_glTF	String, 120	Path of glTF File	N
Format_MAX	String, 120	Path of MAX File	N

3.3.3. Building Shell – 3D Visualisation Map (Tile-based Models) (Grid Index)

3.3.3.1. Tile_based_models

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
SHEETNO	String, 9	iB1000 Sheet No	N
URL_3D_Visualisation_Map	String, 25	URL of 3D Visualisation Map	N
Download_API	String, 35	URL of Download API	N
REVISIONDATE	Date	Revision Date	N
Format_OBJ	String, 120	Path of OBJ File	N
Format_OSGB	String, 120	Path of OSGB File	N
Format_3D_Tiles	String, 120	Path of Cesium 3D Tiles File	N

3.3.4. Building Shell – 3D Visualisation Map (Non-textured Models) (Grid Index)

3.3.4.1. Non_textured_models

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
SHEETNO	String, 9	iB1000 Sheet No	N
URL_3D_Visualisation_Map	String, 25	URL of 3D Visualisation Map	N
Download_API	String, 35	URL of Download API	N
REVISIONDATE	Date	Revision Date	N
Format_FBX	String, 120	Path of FBX File	N
Format_glTF	String, 120	Path of glTF File	N
Format_MAX	String, 120	Path of MAX File	N

3.3.5. Floor – Floor

3.3.5.1. Floor

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
FloorID	String, 36	The FloorID must be a globally unique, RFC4122 compliant version 4 Universally Unique Identifier (UUID). It is a computer-generated unique identifier for floor feature.	N
FloorPolyID	String, 28	Floor Polygon Identity Number.	N
ModelID3DSD	String, 18	3D Spatial Data Identity Number of the building at which the floor located. Remarks: If the Floor Polygon located with more than one ModelID3DSD 3D building models, select the one with its BuildingCSUID with the largest area of the building polygon.	N
BuildingCSUID	String, 19	Building Common Spatial Unit Identifier (CSUID) established under the Data Alignment Measures (DAM). Remarks: If the Floor Polygon located with more than one BuildingCSUID building polygons, select the BuildingCSUID with the largest area of the building polygon.	N
BuildingCSUIDCount	Short, 1	Indicate if the Floor Polygon is located with more than one BuildingCSUID building polygons. 1 = Intersect with one BuildingCSUID building polygon 2 = Intersect with two BuildingCSUID building polygons …	N
FloorSubtype	Short, 2	Floor type.	Y
EnglishDisplayFloorName	String, 150	Annotation of the floor level (English) (E.g. G/F, 1/F, B/2...)	Y
ChineseDisplayFloorName	String, 150	The annotation of the floor level (Chinese) (E.g. 地面, 一樓, 地下二層 ...).	Y
FloorSubdivision	String, 2	The divided polygons with different levels but under the same floor. (E.g. 00 = Floor X main part, 01 = Floor X subdivision part 1,…).	Y

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
BaseLevel	Double (3,2)	Reduced level of the Floor Polygon. (in Metres above the Hong Kong Principal Datum (HKPD)).	Y
CreationDate	Date	Creation date of the Floor Polygon.	Y
Creation_by	Short, 2	The organisation that created the Floor Polygon.	Y
LastAmendmentDate	Date	Latest amendment date of the Floor Polygon.	Y
LastAmendment_by	Short, 2	The organisation that amended the Floor Polygon.	Y
BuildingPlanNo	String, 150	Drawing number of the corresponding building plan.	Y
BuildingPlanNoDate	Date	Date of the drawing number of the corresponding building plan.	Y
BuildingPlanSource	Short, 2	Indicate from which the organization the building plan is.	Y
Restricted	String, 1	Indicate whether the Floor Polygon is restricted to access or not Y=Yes N=No	Y

3.3.6. Room – Room

3.3.6.1. Room

Entity type: Feature Layer

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
RoomID	String, 36	The RoomID must be a globally unique, RFC4122 compliant version 4 UUID. It is a computer-generated unique identifier for room features.	N
RoomPolyID	String, 32	Room Polygon Identity Number.	N
FloorPolyID	String, 28	Floor Polygon Identity Number to which the unit belongs.	N
TrueModelID3DSD	String, 18	Room polygon located exactly on the corresponding 3D Spatial Data Identity Number.	Y
TrueBuildingCSUID	String, 19	Room polygon located exactly on the corresponding Building CSUID.	Y
RoomSubtype	String, 5	Room Type.	Y
OmniClassNumber	String, 11	Unique number assigned to an entry under the classification system for the	Y

Field Name	Data Type and Length	Description	Allow Null Value (Y/N)
		construction industry called the OmniClass Construction Classification System (known as OmniClass™ or "OCCS") issued by The OmniClass Construction Classification System (OCCS) Development Committee (http://www.omniclass.org)	
RoomNumberName	String, 150	Annotation of the room number or shop number. (E.g. 201, 201A, 202B...)	Y
EnglishDisplayName	String, 150	Name of the room stated on building plans in English. (E.g. Meeting Room, Elevator...)	Y
ChineseDisplayName	String, 150	Name of the room stated on building plans in Chinese. (E.g. 會議室，電梯 ...)	Y
BaseLevel	Double (3,2)	The bottom level of the Room Polygon. (metres above the Hong Kong Principal Datum (HKPD)).	Y
CreationDate	Date	The creation date of the Room Polygon.	Y
Creation_by	Short, 2	The organisation that created the Room Polygon.	Y
LastAmendmentDate	Date	The latest amendment date of the Room Polygon.	Y
LastAmendment_by	Short, 2	The organisation that amended the Room Polygon.	Y
FieldCheckingDate	Date	The latest onsite field checking date.	Y
FieldCheckingMethod	Short, 2	The method applied for validation and amendment.	Y
Restricted	String, 1	Indicate whether the Room Polygon is restricted or not Y=Yes N=No	Y

3.4. Unique Identifiers

The following table describes the unique identifier for each entity in the Building FSDT.

Data Layer	Unique Identifier	Description
Building	BuildingCSUID	A unique identifier maintained by LandsD, which consists of 19 characters and 3 segments, for example, "3644815271T20030415". The following describes the content of each segment: • "3644815271": Geo-reference No • "T": Polygon Type (T means Tower and P means Podium) • 20030415: Record Creation Date
BuildingWorksHistory	BuildingCSUID, DataProviderID, WorksType, WorksDate	BuildingCSUID and WorksType together form a composite identifier maintained by LandsD, for example, BuildingCSUID "3644815271T20030415" with WorksType "Issue date for the Certificate of Exemption in respect of Site Formation Works" is unique in the data layer.
BuildingName	BuildingNameID, DataProviderID	BuildingNameID and DataProviderID together form a composite identifier, for example, BuildingNameID "1109618882" with DataProviderID "LandsD" is unique in the data layer.
BuildingRelateOPStructure	BuildingCSUID, BuildingStructureID	BuildingCSUID and BuildingStructureID together form a composite identifier, for example, BuildingCSUID "0312012825T20080107" with BuildingStructureID "5246756" is unique in the data layer.
OPStructure	BuildingStructureID	A unique identifier maintained by BD with reference to the concerned Building Block ID as in BD's Building Condition Information System ("BCIS"), for example: "11967336".
OP	OPNo	A unique identifier maintained by BD for Occupation Permit, for example: "K21/94(RU)".
BuildingLotNoInfo	BuildingStructureID	A unique identifier maintained by BD with reference to the concerned Building Block ID as in BD's Building Condition Information System ("BCIS"), for example: "11967336".

Data Layer	Unique Identifier	Description
Individualised_models	SHEETNO	A unique identifier maintained by LandsD which based on the iB1000 map sheet index, consists of four segments, for example, "10-NE-14C". The following describes the content of each segment: • "10": One to two digit(s), no padding zero • "NE": Two letters denoting the four quadrants that the grid lines in, for example, NW for Northwest, NE for Northeast, SW for Southwest and SE for Southeast • "14": One to two digit(s), no padding zero • "C": One letter between A-D, denoting the subdivided grid number
Tile_based_models	SHEETNO	
Non_textured_models	SHEETNO	
Floor	FloorID	A system generated global unique RFC4122 compliant version 4 UUID maintained by LandsD. The ID should be formatted as: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX
	FloorPolyID	A unique Floor Polygon Identity Number maintained by LandsD, which consists of 28 characters and 5 segments, for example, "F1111199999T2000043002100102". The following describes the content of each segment: • "F": Polygon Type, 1 character, F = Floor Polygon • "1111199999T20000430": BuildingCSUID, 19 characters • "02": Subtype of the Floor Polygon, 2 characters. It is the subtype of the Floor Polygon to define each floor from the building plan. Refer to 3D Indoor Maps Specifications provided in the CSDI Portal for the code values. • "1001": Floor Number, 4 characters. It represents the floor level on each floor. Floor Number as an indicator of +ve / -ve sign of Ordinal: • 0 = +ve, 1 = -ve

Data Layer	Unique Identifier	Description
		001: 3 digits of the Ordinal (Z Value) Examples: 1002 = Basement 2 1001 = Lower Ground / Basement 0000 = Ground Floor 0001 = Level 1 0002 = Level 2 … For the skipped floors like 4F, 14F…, the floor numbers will be assigned in ascending order. E.g. the 5F will be 0004 when 4F is not available. Split floors should have separate floor numbers. • "02": Subdivision, 2 characters. For flat roof, upper roof, basement, mezzanine of building and MTR entrance/exit only, floor subdivision applies to the separated parts within the same floor. The separated parts within the same floor are represented as: 00 = Floor X main part 01 = Floor X subdivision part 1 02 = Floor X subdivision part 2 …
Room	RoomID	A system generated global unique RFC4122 compliant version 4 Universally Unique Identifier (UUID) maintained by LandsD. The ID should be formatted as: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX
	RoomPolyID	A unique Room Polygon Identity Number maintained by LandsD, which consists of 32 characters and 3 segments, for example, "U1111199999T200004300210010020001". The following describes the content of each segment: • "U": Polygon Type, 1 character • "1111199999T2000043002100102": FloorPolyID, 27 characters. • "0001": Room Number, 4 characters. It represents the unique number assigned sequentially to each room of the corresponding floor, (i.e. 0001, 0002,…,9999). Thereafter, every new room on the same floor was allocated the

Data Layer	Unique Identifier	Description
		next number sequentially, irrespective of its location.

3.5. Geographical Coverage

The dataset(s) specified in this document covers the entire territories of HKSAR.

3.6. Units of Measurement

The units of measurement used in all datasets under Building FSDT refer to the International System of Units ("SI").

3.7. Spatial Resolution/Scale

The spatial resolution(s) or scale(s) of the data specified in this document is/are:

- Building: The Building entity is maintained at a scale of 1: 1000 to show the approximate location of the Buildings.
- 3D Visualisation Map (Individualised models), 3D Visualisation Map (Tile-based models) and 3D Visualisation Map (Non-textured models): The dataset is maintained with a horizontal accuracy of +/- 0.3m and vertical accuracy of +/- 0.5m to show the approximate location of the mesh models.
- 3D Visualisation Map (Individualised models) (Grid Index), 3D Visualisation Map (Tile-based models) (Grid Index) and 3D Visualisation Map (Non-textured-models): The datasets are maintained at a scale of 1: 1000 to show the location of the grids.
- Floor and Room: The datasets are maintained with horizontal and vertical accuracy of +/- 1.0 m to show the approximate location of the Floors and Rooms.

3.8. Spatial Representation Type

The spatial representation type(s) of the data specified in this document is/are:

- The Building entity is represented as polygons to represent the geographical location of the entity.
- The 3D Visualisation Map (Individualised models), 3D Visualisation Map (Tile-based models)

and 3D Visualisation Map (Non-textured-models) dataset are represented as mesh models to represent the geographical location of the entity.

- The grid indexes of the 3D Visualisation Map (Individualised models), 3D Visualisation Map (Tile-based models) and 3D Visualisation Map (Non-textured-models) are represented as polygons with standard dimension to represent the geographical location of the entity.
- The Floor and Room entities are represented as polygons to represent the geographical location of the entity.

3.9. Spatial Reference System

The spatial data covered in this document are geo-referenced or projected to one of the Coordinate Reference Systems as specified in the Simplified Data Specifications for Coordinate Reference System of the CSDI.

3.10. Relationships and Dependencies with Other FSDTs

The Building FSDT is related to the Address FSDT. The relationship between the Building entity in this FSDT and the BuildingCSU entity in Address FSDT is identified, where the BuildingCSUID are found in both FSDTs, as illustrated in below Figure 3-2

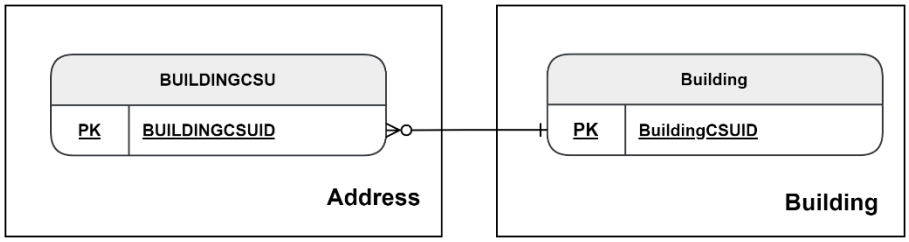


Figure 3-2 - The relationship between Building and Address FSDTs

The English version shall prevail in case of any discrepancy or inconsistency between the English version and its Chinese translation.